Supplementary Materials

S1. Extended Seed Count Robustness

The primary manuscript reports the frozen 2026-05-21 statistics. To evaluate robustness to larger seed counts, we repeated the online benchmark with 16 seeds per condition. This run pooled thalamus, distributed, and distributed meta-monitor control conditions, so it should be interpreted as a supplementary robustness check rather than as a replacement for the primary validation table.

All five constructs remained above the validation threshold (`r >= 0.70`), although the pooled estimates were lower than the primary frozen estimates for several constructs.

Construct	Primary frozen r	16-seed supplementary r	Change
Action agency	0.874	0.727	-0.147
Boundary self	0.940	0.918	-0.022
Identity-temporal self	0.850	0.749	-0.101
Action ownership	0.996	0.987	-0.009
Distributed body-schema self	0.924	0.902	-0.022

The largest change was observed for action agency. The 16-seed estimate (`r=0.727`, 95% CI `[0.662, 0.781]`) is close to the validation threshold but remains positive, statistically reliable, and above criterion. We interpret this as evidence that the primary 8-seed estimate was somewhat upward-biased, not as evidence against the construct. The action-loop mechanism effect also replicated in the extended run (`+0.698`, 95% CI `[0.695, 0.700]`).

S2. Minimal Attention-Based Extended Validation

We also repeated the minimal attention-based 2x2 follow-up with 16 seeds per condition. The result reinforces the interpretation in the main manuscript: action-loop effects generalize to an attention-based substrate, while self-related validation in the minimal attention implementation remains architecture-limited.

Construct	n	r [95% CI]	Interpretation
Action agency	64	0.999826 [0.999734, 0.999907]	near-ceiling convergence
Boundary self	64	0.489 [0.286, 0.674]	not validated
Identity-temporal self	64	0.610 [0.444, 0.747]	exploratory
Action ownership	64	0.999992 [0.999989, 0.999996]	near-ceiling convergence

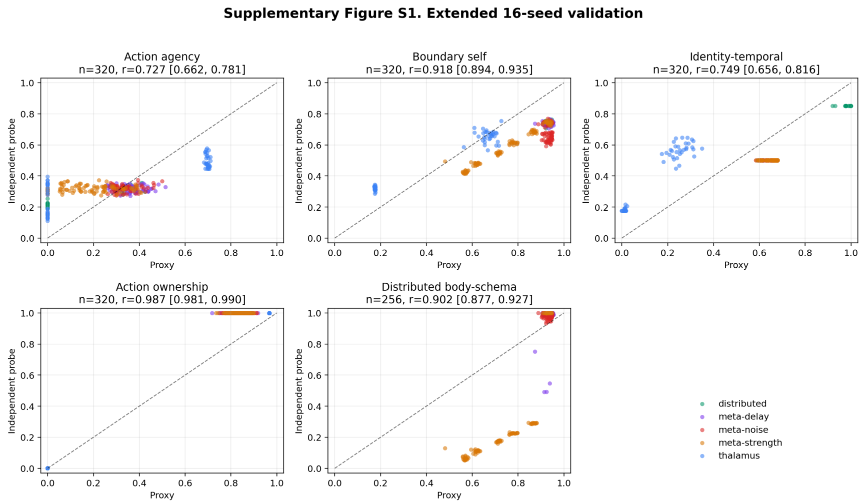

Figure 1: Extended 16-seed validation.

The mechanism effects remained selective. Action-loop availability increased action agency (`+0.8190`, 95% CI `[0.8182, 0.8198]`) and action ownership (`+0.8190`, 95% CI `[0.8182, 0.8197]`). Workspace availability strongly increased the internal boundary-self proxy (`+0.8357`, 95% CI `[0.8287, 0.8426]`) but did not yield validated boundary-self convergence. Identity-temporal self also dropped into the exploratory range in this extended run. This supports a conservative conclusion: the minimal attention system reproduces the action-loop mechanism pattern, but current self-related probes do not provide robust construct validation in this substrate.

S3. Workspace Capacity Sweeps

We ran workspace capacity sweeps in the thalamus-inspired and minimal attention-based architectures. Each sweep tested capacities `0, 1, 2, 4, 8` with 16 seeds per capacity. The contrast between the two architectures clarifies the meaning of the attention-substrate measurement limitation.

S3.1 Thalamus Architecture

In the thalamus-inspired architecture, workspace capacity produced graded changes in workspace-linked measures. Proxy-independent convergence remained strong for both boundary self and identity-temporal self:

Diagnostic	n	Pearson r [95% CI]	Spearman rho
capacity vs boundary proxy	80	0.668 [0.588, 0.737]	0.826

Diagnostic	n	Pearson r [95% CI]	Spearman rho
capacity vs boundary independent	80	0.391 [0.222, 0.535]	0.343
capacity vs identity proxy	80	0.417 [0.262, 0.543]	0.271
capacity vs identity independent	80	0.508 [0.395, 0.608]	0.442
boundary proxy vs independent	80	0.838 [0.722, 0.910]	0.325
identity proxy vs independent	80	0.982 [0.965, 0.992]	0.629

S3.2 Minimal Attention-Based Architecture

In the minimal attention-based architecture, workspace capacity affected the internal boundary proxy but did not rescue independent boundary validation:

Diagnostic	n	Pearson r [95% CI]	Spearman rho
capacity vs boundary proxy	80	0.494 [0.371, 0.596]	0.203
capacity vs generic boundary probe	80	-0.257 [-0.463, -0.034]	-0.078
capacity vs hard attention boundary	80	0.184 [-0.039, 0.386]	0.251
proxy vs generic boundary	80	0.404 [0.208, 0.578]	0.483
proxy vs hard attention boundary	80	0.252 [0.077, 0.435]	0.360

S3.3 Architecture-Conditioned Interpretation

The capacity sweeps show that workspace capacity is not sufficient, by itself, to establish boundary-self validity in every architecture. In the thalamus-inspired architecture, capacity-linked proxy changes corresponded to strong proxy-independent convergence. In the minimal attention-based architecture, capacity changed internal workspace diagnostics but did not produce validated boundary-maintenance behavior. This pattern supports the main manuscript's architecture-conditioned interpretation: mechanisms may generalize while construct validity remains architecture dependent.

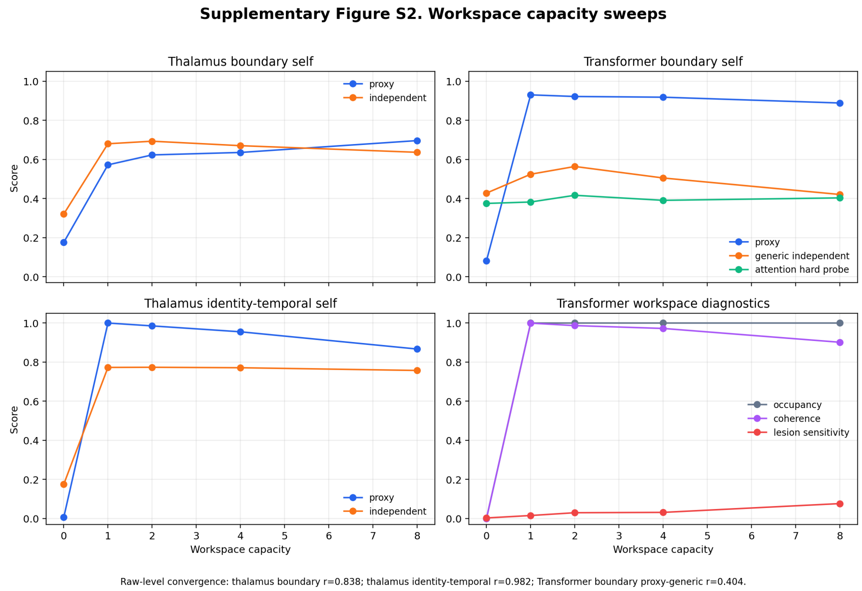

Figure 2: Workspace capacity sweeps.

S4. Independent Test Operational Definitions

The main manuscript summarizes the independent tests used to validate each operational construct. This table gives the corresponding input, operation, and output metric.

Test	Input	Operation	Output metric
Intentional-binding analogue	self-generated actions and delayed effects	compare perceived or represented action-effect interval under controllable versus externally imposed action conditions	temporal compression score
Error-attribution test	action predictions, observed effects, and injected mismatches	ask whether mismatch is attributed to self-generated action or external perturbation	attribution accuracy

Test	Input	Operation	Output metric
Controllability-preference test	paired environments with different action-outcome reliability	measure preference or score advantage for the controllable environment	controllability preference
Forced-action probe	observations with actions selected externally	compare agency score under chosen versus forced actions	forced-action agency drop
Boundary perturbation	self-marked and non-self-marked state components	perturb each component class and measure recovery or protection	self-boundary recovery advantage
Self/other discrimination	own traces and matched traces from other systems	classify or choose which trace belongs to self	discrimination accuracy
Computational mirror test	delayed or transformed prior behavior traces	recognize own trace after delay or transformation	self-recognition accuracy
Delayed identity recognition	identity markers after intervening dynamics	recover or recognize the initial identity marker	marker-recognition score
Temporal-binding probe	shuffled or separated event sequences	reorder or bind events into a coherent temporal sequence	ordering/binding accuracy
Forced-choice ownership	one self-generated action and one external action	choose which action was self-generated	forced-choice accuracy
Ownership-illusion resistance	externally generated outcomes that conflict with own action prediction	reject externally caused outcomes as self-owned	illusion-resistance score

Test	Input	Operation	Output metric
Meta-monitor lesion	distributed system before and after disabling meta-monitoring	compare body-schema and coordination scores under lesion	lesion sensitivity
Hidden-agent perturbation	subtly damaged agent inside a distributed system	detect and respond to the damaged agent	detection and response score
Body-schema update probe	added, removed, swapped, or sensor-modified agents	update the internal body schema after configuration change	convergence speed and final accuracy

S5. Minimal Attention Diagnostic Stress Tests and Construct Variants

The main manuscript reports the compact interpretation of the minimal attention follow-up. The detailed diagnostic statistics are listed here.

S5.1 Near-Ceiling Correlations and Difficulty Diagnostics

The low-noise minimal attention 2x2 run produced near-ceiling proxy-test convergence for action agency and action ownership:

Construct	n	r [95% CI]	Interpretation
Action agency	32	0.999825 [0.999677, 0.999947]	near-ceiling convergence
Action ownership	32	0.999995 [0.999990, 0.999999]	near-ceiling convergence

When the diagnostic sweep varied action-effect noise and action-loop learning rate, convergence became less stable:

Diagnostic subset	Construct	n	r [95% CI]
all diagnostic conditions	action agency	104	0.795 [0.762, 0.836]
all diagnostic conditions	action ownership	104	0.664 [0.589, 0.739]

Diagnostic subset	Construct	n	r [95% CI]
noise sweep only	action agency	48	0.991 [0.986, 0.994]
learning-rate sweep only	action agency	40	0.266 [0.030, 0.516]
learning-rate sweep only	action ownership	40	-0.122 [-0.442, 0.225]

These diagnostics indicate that the near-ceiling values in the low-noise run should be treated as over-convergent rather than as unusually strong independent validation.

S5.2 Minimal Attention Boundary Probes

Workspace dose tracked the internal boundary proxy but did not validate against independent boundary tests:

Diagnostic	n	r [95% CI]
workspace dose vs internal boundary proxy	40	0.728 [0.564, 0.845]
proxy vs generic boundary probe	40	0.325 [-0.023, 0.620]
proxy vs attention-specific hard boundary probe	40	0.246 [-0.023, 0.520]
workspace dose vs attention-specific hard boundary probe	40	0.223 [-0.049, 0.514]

This supports the main text's measurement-limited interpretation for attention-based boundary self.

S5.3 Minimal Attention Construct-Variant Mini-Study

Two intuitive attention-specific candidates did not validate:

Candidate	Proxy-test r	Interpretation
Attention-focus self	0.203	not validated

Candidate	Proxy-test r	Interpretation
Context-window self	0.152	not validated
Context-window dose vs independent test	0.341	weak dose signal only

The more informative pattern was a high-agency / low-boundary profile. Across the agency-boundary grid, `36/96` rows (`37.5%`) met the criterion of high independent agency (`>=0.70`) and low independent boundary score (`<0.50`). Independent agency and independent boundary scores were largely dissociated (`r=0.129`). This is retained as an architecture-conditioned diagnostic profile rather than a validated construct.

S6. Data Availability

Supplementary statistics and run metadata are stored in `docs/paper/statistics/supplementary_20260522/`. Supplementary figures are stored in `docs/paper/figures/supplementary_20260522/`.

The source run directories are:

- `runs/benchmark_online_supplementary/online_20260522_173251/`
- `runs/supplementary/transformer_validation_20260522_173251/`
- `runs/supplementary/transformer_workspace_capacity_20260522_173251/`
- `runs/supplementary/thalamus_workspace_capacity_20260522_173251/`

These supplementary runs are not used to overwrite the frozen primary manuscript statistics.